CV

Maria Dimakopoulou

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EDUCATION

STANFORD UNIVERSITY **2015 – 2018,**
Ph.D. in Reinforcement Learning & Causal Inference **Stanford, CA**
Advisors: Benjamin Van Roy & Susan Athey
GPA 4.26/4.30

- Recipient of Stanford's "Outstanding Academic Achievement at the Doctoral Level" Award.
- Recipient of the "Arvanitidis in Memory of William K. Linvill" Stanford Graduate Fellowship.

STANFORD UNIVERSITY **2015 – 2016,**
M.Sc. in Operations Research **Stanford, CA**
GPA 4.23/4.30

- Recipient of Stanford's "Outstanding Academic Achievement at the Masters Level" Award.
- Graduated 1st of the class.

NATIONAL TECHNICAL UNIVERSITY OF ATHENS (NTUA) **2009 – 2014,**
M.Sc. and B.Sc. in Electrical Engineering & Computer Science **Athens, Greece**
GPA 10.00/10.00

- Graduated with highest GPA in 200-year history of National Technical University of Athens.
- Nationwide 1st in the NTUA Electrical & Computer Engineering department entry exams.

PROFESSIONAL EXPERIENCE

UBER
Mobility Marketplace

- **Director of Engineering – Head of Marketplace Planning & Intelligence (3/2025 – present)**
  Leading the 100+ person Marketplace Planning & Intelligence organization (Engineering & Science) at Uber, which is responsible for the algorithms and engineering systems determining rider & driver pricing & incentive levels for Uber's Mobility Marketplace, as well as for strategic Marketplace data & metrics across Mobility & Delivery. The Marketplace Planning & Intelligence organization serves as the interface between Uber's Technology and Central Operations & Finance departments, driving innovation across Deep Learning for baseline forecasting, Causal Inference & Causal ML for effect estimation, Operations Research for large-scale optimization, Reinforcement Learning for long-term decision making, and Data Science, Backend, ML Infrastructure & Data Engineering at scale.

SPOTIFY
Personalization Mission

- **Director of Machine Learning Engineering – Head of Homepage Personalization (11/2022 – 2/2025)**
  Led the 60+ person Homepage Personalization organization that was responsible for the personalized recommendations across music, podcasts and audiobooks on the Homepage of 650+ million Spotify users. Drove strategy & execution across Machine Learning, Backend and Data at scale, with the goal to serve recommendations that maximize user satisfaction, while enabling multiple strategic & business objectives on the Spotify homepage.

- **Director of Machine Learning Engineering – Spotify Bet Strategy Lead (9/2024 – 2/2025)**
  Concurrently to my Head of Homepage Personalization role, I served as the company-wide Strategy Lead of Spotify's "Programming for Distribution" Bet, involving 120+ FTEs across Engineering, Product Management, Data Science & Design and focused on enhancing content discovery for Spotify's listeners.
- **Senior Manager of Machine Learning Engineering – Head of Home Assembly & Ranking (1/2022 – 11/2022)**
  Led two teams within the Homepage Personalization organization at Spotify; one responsible for developing a unified relevance ranker for all Spotify content and the other responsible for optimizing the personalized placement of organic and promotional content on the homepage, to balance user satisfaction and Spotify's business needs.

NETFLIX RESEARCH
Product ML Research, Data Science & Engineering

- **Adaptive Experimentation Tech Lead (1/2020 – 1/2022)**
  Built a cross-functional working group of researchers, data scientists and engineers to deliver new experimentation capabilities for Netflix as an alternative to A/B testing that increased experimentation velocity & rollout decisions. (1) automated ML-based A/B test cell selection, (2) adaptive cell allocation rates during-test, (3) valid inference from adaptively collected data post-test, (4) ML-based short-term proxy metrics of long-term A/B test metrics.
- **Causal Machine Learning Research Area Lead (12/2018 – 1/2022)**
  Pioneered the causal ML investment at Netflix and established its usage across Messaging personalization, Netflix Billboard personalization and UI personalization. Drove Netflix Research's initiatives in the intersection of Causal Inference & Machine Learning, designing algorithms and driving their adoption in production. Authored internal methodology guide, co-authored several papers accepted in top-tier ML conferences and served as co-organizer or invited speaker in multiple internal forums and external venues.

MICROSOFT RESEARCH
NYC Machine Learning Lab

- **Visiting Research Scientist (6/2018 – 9/2018):** Designed a reinforcement learning value-function decomposition allowing the breakdown of complex RL tasks into simpler ones that can be learned faster by multiple RL agents (in partnership with Miro Dudik, Rob Schapire). Designed a range of new off-policy estimators that exhibit better bias-variance properties than existing ones in slate contextual bandit settings (in partnership with Miro Dudik)**.**

SALESFORCE
Data Science Team

- **Visiting Research Scientist (6/2016 – 9/2016):** Designed and implemented end-to-end the Salesforce DMP (f.k.a. Krux) Multi-Touch Attribution algorithm. Launched it to a group of high-profile clients. Featured in Mobile Marketing Association's review for its 800 member-companies.

GOOGLE
Google Research

- **Ad Exchange Optimization Team Software Engineering Intern (11/2014 – 4/2015)**: Developed a dynamic pricing algorithm that yielded annual revenue lift for Google's Real-Time Bidding in the scale of hundreds of millions of dollars. Launched it in the production Ad Exchange Dynamic Pricing Pipeline, with praise from the VP of Google Ad Exchange.
- **Operations Research & Optimization Software Engineering Intern (7/2013 – 11/2013):** Designed algorithm that solved the 3-year unsolved measurement corruption erratum on Intel processors impacting 1000s of Intel machines. Improved scheduling of hardware events on the processors' counters in the Google production kernel

that increased measurement accuracy and hardware utilization by 18%.

- **Operations Research & Optimization Software Engineering Intern (8/2012 – 11/2012):** Designed linear & mixed integer programming models for convex & non-convex cost optimization scheduling problems in Google Technical Infrastructure and Google Geo.

PUBLICATIONS & PREPRINTS

- **Concurrent Learning with Aggregated States via Randomized Least Squares Value Iteration**
  Chen, Bai, Zhang, Dong, Dimakopoulou, Sun, Zhou – *ICML 2025*
- **Calibrated Recommendations with Contextual Bandits**
  Feijer, Abdollahpouri, Gupta, Clare, Wen, Wasson, Dimakopoulou, Nazari, Kretschman, Lalmas – *arXiv:2509.05460*
- **Evaluating the Surrogate Index as a Decision-Making Tool Using 200 A/B Tests at Netflix**
  Zhang, Zhao, Le, Dimakopoulou, Kallus – *arXiv:2311.11922; MIT CODE 2023*
- **Calibrated Recommendations as a Maximum Flow Problem**
  Abdollahpouri, Nazari, Gain, Gibson, Anderton, Dimakopoulou, Carterette, Lalmas, Jebara – *WSDM 2023*
- **Society of Agents: Regrets Bounds of Concurrent Reinforcement Learning**
  Chen, Dong, Bai, Dimakopoulou, Xu, Zhou – *NeurIPS 2022*
- **Online Multi-Armed Bandits with Adaptive Inference**
  Dimakopoulou, Ren, Zhou – *NeurIPS 2021*
- **Post-Contextual-Bandit Inference**
  Bibaut, Dimakopoulou, Kallus, Chambaz, van der Laan – *NeurIPS 2021*
- **Risk Minimization from Adaptively Collected Data: Guarantees for Supervised and Policy Learning**
  Bibaut, Kallus, Dimakopoulou, Chambaz, van der Laan – *NeurIPS 2021*
- **Sequential Causal Inference in a Single World of Connected Units**
  Bibaut, Petersen, Vlassis, Dimakopoulou, van der Laan – *arXiv:2101.07380*
- **Doubly Robust Off-Policy Evaluation with Shrinkage**
  Su, Dimakopoulou, Krishnamurthy, Dudik – *ICML 2020; US Patent 16/657,533*
- **ADMM SLIM: Sparse Recommendations for Many Users**
  Steck, Dimakopoulou, Riabov, Jebara – *WSDM 2020*
- **Marginal Posterior Sampling for Slate Bandits**
  Dimakopoulou, Vlassis, Jebara – *IJCAI 2019*
- **On the Design of Estimators for Bandit Off-Policy Evaluation**
  Vlassis, Bibaut, Dimakopoulou, Jebara – *ICML 2019*
- **Balanced Linear Contextual Bandits**
  Dimakopoulou, Zhou, Athey, Imbens – *AAAI 2019*
- **Scalable Coordinated Exploration in Concurrent Reinforcement Learning**
  Dimakopoulou, Osband, Van Roy – *NeurIPS 2018*
- **Coordinated Exploration in Concurrent Reinforcement Learning**
  Dimakopoulou, Van Roy – *ICML 2018*
- **Estimation Considerations in Contextual Bandits**
  Dimakopoulou, Zhou, Athey, Imbens – *arXiv:1711.07077; NeurIPS 2017 ML & Causal Inference workshop*

- **Market-Based Dynamic Service Mode Switching in Virtualized Wireless Networks**
  Dimakopoulou, Bambos, Valdez-Vivas, Apostolopoulos – *IEEE PIMRC 2017*
- **Reliable and Efficient Performance Monitoring in Linux**
  Dimakopoulou, Eranian, Koziris, Bambos – *ACM/IEEE Supercomputing 2016*

HONORS & AWARDS

- **Greek International Women Awards (2025)**
  for worldwide outstanding professional achievement in the field of Science & Technology.
- **Spotify 20/20 R&D Senior Leadership Program (2022)**
  selected in Spotify's R&D senior leadership 2022 cohort.
- **Stanford University Outstanding Academic Achievement at the Doctoral Level Award (2019)**
  presented annually to the top graduating Ph.D. student of Stanford, MS&E.
- **Forbes 30 Under 30 Greece (2019)**
  presented to "the brightest young entrepreneurs, leaders, and stars of Greek nationality worldwide".
- **Stanford University "Arvanitidis in Memory of William K. Linvill" Graduate Fellowship (2015 – 2018)**
  awarded for excellence in research and study.
- **Onassis Foundation Graduate Fellowship (2015 – 2018)**
  awarded annually to the best performing Ph.D. students of Greek nationality worldwide.
- **Stanford University Outstanding Academic Achievement at the Masters Level Award (2016)**
  presented annually to the top graduating M.Sc. student of Stanford, MS&E.
- **National Technical University of Athens Dean's Council "Papakyriakopoulos" Honorary Award (2015)**
  presented annually to the best performing student in Mathematics coursework across all NTUA.
- **Limmat Stiftung Foundation Academic Excellence Award (2015)**
- **Intel Honorary Award (2014)**
  presented "in recognition for the creativity and drive in modifying the Linux Performance Monitoring Subsystem to improve the PMU accuracy in 1000s of Intel machines".
- **Google Women Techmakers – Anita Borg Memorial Fellowship (2014)**
  presented "for excellence in computer science and technology, outstanding academic achievement, leadership and community involvement".
- **Google Spot Bonus Excellence Award (2013)**
  presented for "multiple accomplishments and outstanding achievements in the Operations Research team".
- **Google Peer Bonus Volunteering Award (2013)**
  for "promoting careers in engineering to high-school girls and assisting Google's diversity efforts".
- **National Technical University of Athens Dean's Council "Thomaidion" Honorary Award (2011 – 2015)**
  for excellence in academic performance across all NTUA departments.
- **State Scholarship Foundation of Greece Fellowship (2011 – 2015)**
  for excellence in academic performance in NTUA's Electrical Engineering & Computer Science department.
- **NTUA Dean's Council "Agoniston Polytechniou" Honorary Award (2011)**
  for ranking nationwide 1st in the entry exams of NTUA's Electrical Engineering & Computer Science.
- **Triantafyllidis Foundation, Greek Finance Ministry & Greek Education Ministry Fellowship (2011)**
- **Eurobank EFG & Greek Education Ministry "Great Moment for Education" Honorary Award (2010)**
  for ranking nationwide 1st in the entry exams of NTUA's Electrical Engineering & Computer Science.
- **26th National Mathematics Olympiad Silver Medal (2009)**
- **Hellenic Mathematical Society Silver Medal at the "Euclid" Contest (2008)**
- **6th European Union Science Olympiad Bronze Medal (2008)**

- **Hellenic Mathematical Society Bronze Medal at the "Euclid" Contest (2005 – 2007)**

TALKS & WORKSHOPS

- Invited speaker at **2026 Econometrics and Statistics (EcoSta) International Conference** (August 10th 2026).
- Invited contributor at the **2026 MIT Workshop on Aligning Superintelligence** (May 13th 2026)
- Invited contributor at the **2025 MIT Workshop on Optimizing for the Long-Term** (November 13th 2025)
- Keynote speaker at **Industrial Recommendation Systems Workshop of ACM CIKM 2024** (October 25th 2024).
- Invited speaker at **Bernoulli IMS of the 2024 World Congress in Probability & Statistics** (August 14th 2024).
- Invited speaker at **Intel 2024 MLCon Conference** (June 6th 2024).
- Invited speaker at **Stanford 2024 Causal Science Center Conference on Experimentation** (May 31st 2024).
- Invited speaker at **Netflix 2024 Personalization, Recommendation & Search Annual Summit** (May 31st 2024).
- Invited speaker at **Stanford/UCSF Bay Area Tech Economics Seminar** (December 7th 2023).
- Co-organizer of **RecSys 2022 MORS Workshop on Multi-Objective Recommenders** (September 23rd 2022).
- Co-organizer of **RecSys 2022 REVEAL Workshop on Reinforcement Learning for Recommender Systems** (September 22nd 2022).
- Invited speaker at **Women in Data Science (WiDS) Conference** (March 17th 2022).
- Co-organizer of **NeurIPS 2021 Causal Sequential Decisions Workshop** (December 14th 2021).
- Invited speaker at **Wharton School of the University of Pennsylvania** (October 25th 2021).
- Invited speaker and panelist at **ICML 2021 RL4RealLife Workshop** (July 23rd 2021).
- Invited speaker at **Netflix 2021 Personalization, Ranker and Search (PRS) annual Summit** (June 11th 2021)**.**
- Invited speaker at **NeurIPS 2020 Expo** (December 6th 2020).
- Co-organizer of **RecSys 2020 Bandit and Reinforcement Learning from User Interactions (REVEAL) Workshop** (September 26th 2020)
- Invited speaker at **ICML 2020 Women In Machine Learning workshop** (July 13th 2020).
- Invited speaker at **INFORMS 2019 Bandits and Reinforcement Learning session** (October 20th 2019).
- Co-organizer of **RecSys 2019 Bandit and Reinforcement Learning from User Interactions (REVEAL) Workshop** (September 20th 2019).
- Invited speaker at **DeepMind** (September 16th 2019).
- Speaker at **IJCAI 2019 Online Learning Main Conference Track** (August 14th 2019).
- Speaker at **AAAI 2019 Machine Learning Main Conference Track** (January 29th 2019).
- Invited speaker at **Cornell's School of Operations Research and Information Engineering Young Researcher Workshop** (October 12th 2018).
- Invited speaker at **Google NYC Algorithms Seminar** (July 24th 2018).
- Speaker at **ICML 2018 Reinforcement Learning Main Conference Track** (July 11th 2018).
- Invited speaker at **Netflix Research Seminar** (May 25th 2018).
- Invited speaker at **NeurIPS 2017 From What If to What's Next: ML and Causal Inference Workshop** (December 8th 2017).